

# APM Project Management Qualification

Learner Study Pack



# Quality Management

## Introduction

This section of the Learner Study Pack will cover the Quality Management competence of the APM Project Management Qualification.

The training on this topic will enable you to gain an understanding of the three elements to Quality Management and how they work together to confirm the project is meeting stakeholder requirements and are fit for purpose.

## Learning objective and outcomes

### Learning objective:

18. Understand quality management as the ability to ensure that outputs are delivered in accordance with requirements.

### By the end of this activity, you should be able to:

18a) Understand what is meant by quality planning, including an understanding of quality indicators and how these relate to the business case.

18b) Knowledge of how quality control techniques are used to determine whether success criteria are met.

## Study time

This section of the Learner Study Pack will take approximately 1 - 2 hours to complete.

## Recommended resources

In addition to your Learner Study Pack, use the below references for more information on Quality Management:

- APM Body of Knowledge Seventh edition, 4.1.5 (Quality planning); 4.3.8 (Quality control)
- APM Learning Module, Quality Management. Accessed at: <https://www.apm.org.uk/resources/apm-learning/>
- APM Resources, what is quality management and control? Accessed at: <https://www.apm.org.uk/resources/what-is-project-management/what-is-quality-management-and-control/>
- APM Contracts and procurement Specific Interest Group (2017) Guide to Contracts and Procurement. Princes Risborough: Association for Project Management.

## Overview

Quality planning is the process for establishing quality for a project. It involves taking the defined scope and listing the criteria to be used to validate that the outputs are fit for purpose, and the methods that will be used to do this. This then forms the quality plan which is approved by the sponsor and wider governance board.

The quality plan is a key part of the project management plan (PMP) and sets out the work in scope, aligned to the business case, and how it will be assessed. It includes regulations, standards, specifications, and the agreed acceptance criteria. This then provides guidance to the project team as it answers the question **‘what does good look like?’**

The quality plan guides the planning of quality control and assurance activities that also happen to confirm the outputs meet requirements. All these activities require resources and will influence tasks on the project that need to be estimated and planned into the schedule and cost.

Finally, getting stakeholder agreement for the quality plan supports a successful handover on project completion, by making sure that what is being delivered by the project team achieves the outcomes of the business case and to the required standards.

A Quality Plan includes:

1. Methods of verifying that the outputs meet requirements.
2. Pass / fail criteria for each method.
3. Frequency of the tests, checks or audits that are to be carried out.
4. Requirements for resources required to complete the activity in the quality plan.

## Quality management

- Quality control is verifying compliance of the output to requirements, whereas quality assurance is providing confidence that the right processes and methods have been used to deliver it.
- When using an iterative life cycle, acceptance criteria will be defined for each time period, and then will also be verified for the final product, once the project is delivered, to confirm the acceptance criteria has been met for the overarching outputs.
- If a project is part of a programme, it may be that some of the quality requirements are planned and controlled at programme level.



## Quality control vs quality assurance

Quality control and quality assurance are two distinct components of quality management.

### Quality control

- Consists of inspection, measurement and testing to confirm project outputs meet acceptance criteria defined during quality planning.
- Focused on finding problems to stop them from being passed onto the end customer.
- Can be performed by a member of the project team.
- To be effective, the test plans agreed during quality planning need to be managed through change control, so any amendments are agreed and communicated.

Test plans include aspects such as:

- Sample size of tests
- Test protocols including resources required
- Independent performance or witnessing of tests
- To be commenced when physical outputs are delivered to be tested against.
- Outputs are checked against degree of conformance to the requirements and acceptance criteria. If non-conformance, decisions need to be made on what action is needed.

- Quality control as a process will provide an answer that is either a pass or fail with no in-between.

### **Quality assurance**

- Looks to build in quality using standard processes and procedures supported by training and feedback.
- Looks to maintain stakeholder confidence that a project has been delivered in the right way.
- Is performed by people independent to the project.
- To be effective, a framework with supporting documentation on how to do things and methods to use needs to be in place as standard practice.
- To be applied as soon as the project commences management activity.
- In the event a deviation to standards is identified, a decision will need to be made on what action is needed.
- Quality assurance as a process could not be being followed fully yet still deliver an acceptable output.

**Scenario One:**

As more and more materials begin to arrive onsite for the renovation project to begin, it has become apparent that one of the major challenges will be ensuring the quality of them.

The concrete supplier has just delivered a large batch of concrete and there are questions being raised by the sub-contractors that have been brought in to lay the foundations as to whether it meets the specified strength requirements.

You've decided to go back to the original Quality Management Plan that was set up with the builders and suppliers and are reviewing each of the actions that it outlined.

**Using the 3 steps of Quality Management (Quality Planning, Quality Assurance and Quality Control), what specific actions could have been taken to ensure that substandard concrete was not delivered to site?**

**Quality Planning:**

**Quality Assurance:**

**Quality Control:**

**Scenario Two:**

After reviewing the Quality Management Plan that you put into place before construction began, you can see that all the necessary steps and risks had been identified and agreed to by all suppliers, as well as the builders and sub-contractors too.

**What actions could you take to understand how a substandard batch of concrete was delivered?**

**What kind of documentation could support you in identifying what went wrong?**

**How could you ensure that this scenario does not happen again?**



## Learning review questions

Before you attempt the exam practice questions in the next section, we recommend you review your learning to quickly see which aspects you don't understand or can't remember as well.

It will also boost your confidence by confirming what you have learnt and your demonstrated ability to apply that knowledge, so that you are ready to practise for the exam.

Here are some questions you may wish to answer to review your learning from the Quality Management section:

- What is the difference between Quality Assurance and Quality Control? What activities are associated with each of them?
- Why is it essential to establish quality standards and criteria during the planning phase of any project?
- What is the role of Quality Control in addressing non-compliance or defects discovered during the project?
- How could you ensure that Quality Management practices and principles are integrated into the lifecycle of a project?
- What lessons can be learnt by a project manager from quality incidents or non-compliance situations/issues?
- How does a Quality Management Plan contribute to stakeholder engagement, satisfaction and trust in the delivery of the project?

## Exam practice questions

**18.1** Which three of the following statements are true about quality planning and a quality plan?  
(1 mark)

1. Quality planning involves defining quality objectives and identifying the processes required to deliver products that meet customer requirements.
2. A quality plan indicates measurable parameters used to assess the performance of a product or process against defined quality standards.
3. Quality planning should consider the financial aspects of quality management but does not directly relate to the business case.
4. A quality plan outlines the steps and resources needed to ensure that products or services meet quality requirements.
5. Quality planning should be an ongoing process, continually reviewed and updated throughout the project lifecycle.
6. Quality planning provides the justification for investing in quality planning activities and outlines the expected benefits and returns on investment.

Select the correct combination of the above statements:

- a. 1, 3, 6
- b. 2, 3, 4
- c. 1, 4, 5
- d. 2, 3, 6

**18.2** Which Quality Control techniques are used to determine project success?  
(1 mark)

- a. Auditing processes, Standardising processes, Observing the Project Manager.
- b. Testing of outputs, Re-planning, Surveying stakeholders.
- c. Inspection of outputs, Generating questions, Testing designs.
- d. Inspection of outputs, Surveying stakeholders, Testing outputs.

**18.3** What is one purpose of a quality plan and what are four elements that should be included in a quality plan?  
(5 marks)

**18.4** What are two elements of information that a Quality Plan may include to ensure a project's success criteria are met?  
(2 marks)

## Exam practice answers

| Question | Answer / mark scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18.1     | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 18.2     | D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 18.3     | <p>1 mark for the purpose of a quality plan. For example, but not limited to:</p> <ul style="list-style-type: none"> <li>• It guides the planning of quality control and assurance activities.</li> <li>• It plans how to confirm that project outputs meet requirements.</li> <li>• It sets out the quality indicators / acceptance criteria of outputs, aligned to the business case.</li> <li>• It answers the question 'what does good look like'?</li> <li>• It includes regulations, standards, specifications and acceptance criteria / quality indicators.</li> </ul> <p>1 mark for each correct element included in a quality plan, 4 marks available in total. For example, but not limited to:</p> <ul style="list-style-type: none"> <li>• How it will be verified that the outputs meet requirements / methods of verification.</li> <li>• Pass / fail criteria for each method.</li> <li>• Frequency of the tests to be carried out.</li> <li>• Requirements for resources to carry out testing.</li> <li>• Testing equipment required.</li> </ul> |
| 18.4     | <p>1 mark for each identified element, 2 marks available in total. For example,</p> <ul style="list-style-type: none"> <li>• Pass/fail criteria for each method / each deliverable.</li> <li>• Quality control processes/activities</li> <li>• The frequency of the tests / checks / audits that will be carried out.</li> <li>• What resources are needed to support / carry out quality tests.</li> <li>• Actions / remediations that can be taken in the event of a fail..</li> <li>• Methods of documenting a pass/fail for each method / each deliverable.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |